

# Axon-X Starter Guide

operators who want to use Axon-X today without learning everything at once

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Axon-X · <http://127.0.0.1:4173> · [axon-watch repo](#)

Early bootstrap guide · 2026-07-05 · Read in layers — you do not need the whole system at once.

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# Axon-X Starter Guide

**Version:** early bootstrap (2026-07-05)  
**For:** operators who want to use Axon-X today without learning everything at once  
**Repo:** `axon-watch` · **URL:** `http://127.0.0.1:4173`

## Read this first (60 seconds)

Axon-X is the **next-generation Axon console**. It is real and usable for bounded local work, but it is **not** a full replacement for classic Axon on port **7734** yet.

At this stage, treat Axon-X as:

- a **workspace-scoped operator shell** (runs, signals, conversation, terminal, files)
- backed by three local services you start with one command
- growing by **thin verified slices**, not big-bang parity

You do **not** need to learn the whole architecture to be productive. Start with **one workspace**, **Operator mode**, and the **Command seam**.

## Two apps on your machine

|        | Classic Axon ( <code>axon-local</code> )           | Axon-X ( <code>axon-watch</code> )                          |
|--------|----------------------------------------------------|-------------------------------------------------------------|
| URL    | <code>http://127.0.0.1:7734</code>                 | <code>http://127.0.0.1:4173</code>                          |
| Start  | <code>./start.sh</code> in <code>axon-local</code> | <code>./scripts/dev/up.sh</code> in <code>axon-watch</code> |
| Status | Daily driver                                       | Early rebuild                                               |

They can run at the same time on different ports. Planning docs still live in `axon-local/Plans/Axon-watch/` ; implementation lives in `axon-watch` .

## Start and stop (the only commands you need daily)

```
cd /home/edp/axon-nvme/repos/axon-watch
npm install          # once, or after dependency changes
./scripts/dev/up.sh # start console + control-plane + watch
```

Open: `http://127.0.0.1:4173`

Health check:

```
./scripts/dev/check-health.sh
```

Stop:

```
./scripts/dev/down.sh
```

**Ports:** 4173 (UI), 8787 (control-plane API), 8788 (watch service).

Logs if something fails: `.local/logs/console-web.log` , `control-plane.log` , `axon-watch.log` .

## What you see: five regions

The shell layout is **locked** — regions do not move between modes.

| Region       | What it is for                                                             |
|--------------|----------------------------------------------------------------------------|
| Top bar      | Identity, runtime strip, KAIRO presence chip, <b>Operator / IDE toggle</b> |
| Left sidebar | Workspaces (and in Operator: <b>Attention</b> toggle)                      |
| Center       | Operator: <b>Mission Control</b> · IDE: Monaco editor + terminal dock      |
| Right dock   | Conversation, Command/KAIRO hero, run context                              |
| Status bar   | Watch / run phase / signals glance strip                                   |

## Operator vs IDE — pick one mindset

### Operator mode (default for monitoring)

Use when you want to **watch, decide, and steer** — not edit code in the center.

- Center = **Mission Control** (current run, live feed, STOP/RESUME/APPROVE/REJECT)
- Terminal dock is **collapsed by default** (reopen via chip or bottom strip)
- Left sidebar has **WORKSPACES | ATTENTION**
- Right dock is **conversation-first**

## IDE mode (default for editing)

Use when you want **files, explorer, and editor**.

- Center = Monaco editor + resizable terminal dock (visible by default)
- Left sidebar = workspaces + file explorer tree
- Right dock keeps full run / approvals / signals stack

**Toggle:** top bar **Operator / IDE** switch. Choice persists in session storage.

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## Workspaces — what they mean today

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A **workspace** is a logical context: files, terminal cwd, chat thread, and runs all scope to the selected `workspace_id`.

### Sidebar workspaces (what the UI shows)

The operator sidebar shows **seven mockup IDs**:

- `workspace_smoke` ← **start here** (acceptance / smoke workspace)
- `workspace_recsys`, `workspace_finance`, `workspace_nlp`, `workspace_cv`, `workspace_edge`, `workspace_research`

Selecting a workspace reloads: file tree, terminal session, chat thread, run context.

**Bootstrap rule:** on load, Axon-X picks active run workspace (if sidebar-visible), else `workspace_smoke`, else the first sidebar entry.

### Hidden / API-only workspaces (important)

The API may list **more** workspaces than the sidebar shows, including:

- `workspace_alpha`, `workspace_bootstrap` (test fixtures)
- `workspace_axon_local` → bound to sibling `../axon-local` repo
- `workspace_axon_watch` → bound to the axon-watch repo itself

These bound workspaces work for **terminal, git, files, and API** — but they are **not in the sidebar yet**. Use the API or curl until UI catalog unification lands.

Bindings file: `config/workspace-project-bindings.json`

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## Your first 10 minutes (recommended path)

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1. `./scripts/dev/up.sh`

2. Open `http://127.0.0.1:4173`
3. Confirm `workspace_smoke` is selected (left sidebar)
4. Stay in **Operator** mode
5. Glance at **Mission Control** (center) — idle standby is normal with no active run
6. Open **Conversation** (right dock) — empty until you send a command
7. In **Command** (bottom hero), type: `git status` → Enter
8. Read the agent reply in Conversation
9. Toggle left sidebar to **ATTENTION** — see runs / approvals / signals stack
10. Optional: click **Open terminal** in Mission Control header → run `pwd` in the PTY

You have now exercised: workspace scope, command executor, mission control, attention sidebar, and terminal attachment.

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## Agents and runs — how work actually happens

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Axon-X does not yet mirror every axon-local agent flow. Today:

### Command seam (chat → bounded executor)

Type natural commands in the **Command** hero. Supported intents:

| You type                                   | What happens                                                     |
|--------------------------------------------|------------------------------------------------------------------|
| <code>health</code>                        | Probes control-plane <code>/api/health</code>                    |
| <code>ls</code> or <code>list files</code> | Lists files in workspace root                                    |
| <code>read README.md</code>                | Reads a workspace file                                           |
| <code>git status</code>                    | Runs git in workspace root (real output for bound repos)         |
| <code>resume from review</code>            | Resumes primary <code>review_ready</code> run for this workspace |

Unsupported text gets a helpful capability list back — not a silent failure.

### Runs (explicit execution units)

Runs have phases (`executing`, `awaiting_approval`, `review_ready`, ...) and capability flags (`can_stop`, `can_resume`, `can_approve`, ...).

#### Where to act on runs:

- **Mission Control** (Operator center): STOP, RESUME, COMPLETE, APPROVE, REJECT
- **Attention sidebar** (Operator): APPROVE / REJECT on pending items

- **Status bar / top runtime strip:** phase glance only (not primary controls)

Creating runs via API is supported ( `POST /api/runs` ); the UI surfaces active runs from canonical control-plane state.

## Conversation rehydration

After a hard reload, Conversation reloads the **latest thread for the current workspace**. Messages do not bleed across workspaces.

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## Connecting real projects (TEST-1 slice)

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Thin workspaces normally live under:

```
.local/workspaces/{workspace_id}/
```

**Bound workspaces** map to real directories via `config/workspace-project-bindings.json` :

```
{
  "bindings": {
    "workspace_axon_local": {
      "project_root": "../axon-local",
      "display_name": "axon-local"
    },
    "workspace_axon_watch": {
      "project_root": ".",
      "display_name": "axon-watch"
    }
  }
}
```

When bound:

- Terminal cwd = real project root (no fake isolated folder)
- `git status` in Command = real repo status
- Monaco file host reads/writes the real tree

**Safety:** paths must fall inside `AXON_WATCH_PROJECT_ROOT_ALLOWLIST` (repo root, parent directory, home — configurable).

Verify bindings:

```
curl -s http://127.0.0.1:8787/api/workspaces/workspace_axon_local | python3 -m json.tool
```

Expect `"connection_kind": "project_path"` and a resolved `project_root` .

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## Cross-workspace handoffs (TEST-2 slice)

Record an explicit handoff when work should continue in another workspace:

```
curl -s -X POST http://127.0.0.1:8787/api/workspaces/workspace_smoke/handoffs \
-H 'Content-Type: application/json' \
-d '{
  "target_workspace_id": "workspace_axon_local",
  "task": "Review axon-local after my smoke test",
  "reason": "Cross-repo follow-up"
}' | python3 -m json.tool
```

Response includes:

- persisted **handoff** record ( `handoff_id` , `status: recorded` )
- **target\_workspace\_summary** (connection metadata + run counts)

List handoffs:

```
curl -s http://127.0.0.1:8787/api/workspaces/workspace_smoke/handoffs | python3 -m json.tool
```

**UI note:** handoffs are API-first today; switching workspace in the sidebar is still manual follow-through.

## Hidden features and power-user tips

### 1. Operator terminal is intentionally hidden

Operator mode stores terminal visibility separately from IDE:

- Default: **collapsed**
- Reopen: Mission Control header **Open terminal** chip, or bottom **Terminal dock** · **Show**
- Close: tab bar **×** on the terminal panel

IDE mode keeps terminal visible by default and restores via status bar **TERMINAL**.

### 2. Left sidebar **ATTENTION** auto-focus

In Operator mode, if approvals or interruptive signals exist, the sidebar may default to **ATTENTION** instead of WORKSPACES — so you see blockers first.

### 3. Live refresh (SSE)

The shell listens to `GET /api/live/events` for refresh hints. If EventSource is unavailable, it falls back to visibility-aware polling.



## 4. Pinned workspace memory

Operator workspace selection can persist in session storage — reload may restore your last explicit pick when bootstrap rules allow.

## 5. KAIRO Briefing rhythm

`GET /api/briefing` feeds **Notice** and **Advise** strings into Mission Control (idle) and the KAIRO hero. This is presentation scaffolding — full KAIRO presence (voice, persona, watch rules) is **not** landed yet.

## 6. API catalog vs sidebar catalog

`GET /api/workspaces` is authoritative; sidebar is a trimmed operator view. Do not assume “not in sidebar” means “does not exist”.

## 7. Acceptance gates (for confidence, not daily use)

```
npm run verify:test0    # workspace_smoke acceptance
npm run verify:test1    # real project connection
npm run verify:test2    # workspace handoff slice
npm run verify          # full contract + UI build gate
```

## 8. Bound workspace git from Command

On `workspace_axon_local` (via API scope or future UI):

```
git status
```

runs against the real axon-local tree when bindings and stack are up.

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## What to do at this early stage (honest scope)

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### Do use Axon-X for:

- exploring the new Operator / IDE shell split
- workspace-scoped terminal + file editing (IDE mode)
- bounded commands ( `git status` , file read/list, health probe)
- watching run phase + approvals + signals in Attention
- verifying slices against `workspace_smoke`
- bound-repo work via API-scoped workspaces ( `workspace_axon_local` )

### Do not expect yet:

- full axon-local feature parity (voice, mobile cockpit, full KAIRO rules, watch connectors)

- every workspace in the sidebar (bound repos are API-visible first)
- automatic cross-workspace agent migration (handoff = record + summary, manual switch)
- production deployment / dedicated-server cutover

**When stuck, check:**

1. Is the stack up? `./scripts/dev/check-health.sh`
2. Is the right workspace selected?
3. Operator vs IDE — are you in the mode that matches your task?
4. Classic Axon still available on **7734** for daily-driver gaps

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## Quick API reference (copy/paste)

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```
# Health
curl -s http://127.0.0.1:8787/api/health

# Workspaces
curl -s http://127.0.0.1:8787/api/workspaces | python3 -m json.tool

# Runtime truth
curl -s http://127.0.0.1:8787/api/runtime/summary | python3 -m json.tool

# Briefing (Notice / Advise)
curl -s http://127.0.0.1:8787/api/briefing | python3 -m json.tool

# Post a chat command
curl -s -X POST http://127.0.0.1:8787/api/chat/messages \
  -H 'Content-Type: application/json' \
  -d '{"workspace_id":"workspace_smoke","content":"git status"}' | python3 -m json.tool

# Workspace chat thread (rehydration)
curl -s http://127.0.0.1:8787/api/workspaces/workspace_smoke/chat/thread | python3 -m json.
```

## Learning in layers (don't read everything at once)

| When you want...   | Read...                                           |
|--------------------|---------------------------------------------------|
| Daily usage        | This guide                                        |
| Deeper handbook    | <code>docs/HOW-TO-HANDBOOK.md</code>              |
| Mission Control UI | <code>docs/OPERATOR_MISSION_CONTROL_V1.md</code>  |
| Project bindings   | <code>docs/WORKSPACE_PROJECT_CONNECTION.md</code> |
| Handoffs           | <code>docs/WORKSPACE_HANDOFF.md</code>            |
| Cutover readiness  | <code>docs/AXON_X_CUTOVER_TODO.md</code>          |
| Layout rules       | <code>docs/UI_LAYOUT_LOCK.md</code>               |

## Troubleshooting (short)

| Problem                         | Fix                                                                        |
|---------------------------------|----------------------------------------------------------------------------|
| Port in use                     | <code>./scripts/dev/down.sh</code> then retry <code>up.sh</code>           |
| Blank runtime / briefing        | Control-plane not ready — check <code>.local/logs/control-plane.log</code> |
| Command says unsupported        | See supported command table above                                          |
| Terminal empty / wrong cwd      | Confirm workspace; bound vs isolated root                                  |
| Conversation empty after reload | Reselect workspace; check <code>/chat/thread</code> API                    |
| Sidebar missing axon-local      | Expected — use API workspace until catalog unifies                         |

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